

VISVESVARAYA TECHNOLOGICAL UNIVERSITY
BELAGAVI – 590 018, KARNATAKA



Shri Bhagwan Mahaveer Jain Educational & Cultural Trust ®
JAIN COLLEGE OF ENGINEERING, BELAGAVI



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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

**PRE-FINAL YEAR
(2025 – 2026)**

MINI-PROJECT REPORT

On

“REAL TIME WATER ANALYSIS SYSTEM”

PROJECT GUIDE

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**DEPARTMENT OF ELECTRONICS AND COMMUNICATION
ENGINEERING**

CERTIFICATE

This is to certify that the **Mini-Project Work** entitled “**REAL TIME WATER ANALYSIS SYSTEM**” carried out by **Abhishek Nidavani (2JI23EC003)**, **Arihant Jakkannavar (2JI23EC022)**, **Ananya Shirol (2JI23EC018)** and **Ishwari Siddanagoudar (2JI23EC049)** are bonafide students of **Department of Electronics and Communication Engineering, Jain College of Engineering, Belagavi**, in partial fulfilment for the award of **Bachelor of Engineering** of the **Visvesvaraya Technological University, Belagavi** during the academic year **2024-2025**. It is certified that all corrections/suggestions indicated for project assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed for the Bachelor of Engineering degree.

Prof. Satish Shinde
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DECLARATION

We **Abhishek Nidavani (2JI23EC003), Arihant Jakkannavar (2JI23EC022), Ananya Shirol (2JI23EC018) and Ishwari Siddanagoudar (2JI23EC049)** students of 5th semester **B.E. Electronics & Communication Engineering, Jain College of Engineering, Belagavi** hereby declare that the dissertation entitled **“REAL TIME WATER ANALYSIS SYSTEM”** has been carried out in a batch and submitted in the partial fulfilment of the requirement for the award of Bachelor’s Degree in Electronics & Communication Engineering under Visvesvaraya Technological University, Belagavi during the academic year **2025– 26**.

<u>Name</u>	<u>USN</u>	<u>Signature</u>
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Arihant Jakkannavar	2JI23EC022	_____
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Place : Belagavi

Date :



VISION

To achieve excellence in education and research for developing globally competent , ethically sound electronic and communication engineers

MISSION

1. To provide a conducive environment for learning through structured student centric, teaching learning process
2. To nurture needs of society by infusing scientific temper in students and to grow as a centre of excellence with efficient industry-institute interaction
3. To inculcate self learning skills, entrepreneurial ability and professional ethics.

Program Educational Objectives (PEO's)

Graduates will be able to

- Contemplate real-time social problems and deliver efficient solutions
- Lead and succeed in professional careers
- Contribute through research and entrepreneurship

Program Specific Outcomes (PSO's)

Graduates in the UG program in Electronics and communication engineering will be able to

- Design, verify and develop analog and digital systems by using state of art technology to contribute to the societal needs.
- Apply the knowledge in various domains of IoT, real time systems, communication systems, VLSI and embedded systems, image and signal processing using hardware and software tools.



PROGRAM OUTCOME'S (PO'S)

Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the **engineering and management principles and apply these to one's own work, as a member and leader** in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



Course Outcomes (CO's)

Course Outcome	Course Outcome Statement	Bloom's Level
BEC586.01	Design and develop an electronic system using modern tools by following professional ethics.	L3,L4
BEC586.02	Develop managerial, teamwork and communication skills.	L3
BEC586.03	Understand the design factors to be considered that impact society.	L2,L3

Bloom's Cognitive Levels

L1: Remembering; L2: Understanding; L3: Applying; L4: Analyzing; L5: Evaluating; L6: Creating

STRENGTH OF CO MAPPING TO PO/PSOS WITH JUSTIFICATION:

1. Slight (Low) 2. Moderate (Medium) 3. Substantial (High)

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	3	1	3	-	-	2	-	-	-	2	2	2
CO2	-	-	-	-	-	-	-	-	3	3	2	-	-	-
CO3	-	-	-	-	-	2	2	-	-	-	-	-	-	-

CO – PO - PSO	Justification
CO1 → PO1 (3) CO1 → PO2 (3) CO1 → PO3 (3) CO1 → PO4 (1) CO1 → PO5 (3) CO1 → PO8 (2) CO1 → PO12 (2) CO1 → PSO1(2) CO1 → PSO2(2)	- Students apply the electronics and communication engineering skills and design products to solve complex engineering problems. - Students gain the knowledge of modern tools and skill usage. - Students can use research-based knowledge and methods including design of experiments, analysis and interpretation to provide valid conclusions. - Students can work ethically and professionally in the industry. - Students can design their projects in analog or digital fields. - Students can choose any technology to build their projects.
CO2 → PO9 (3) CO2 → PO10 (3) CO2 → PO11 (2)	- Students can effectively act as an individual, and as a member or leader and work in a team. - Students can comprehend and write effective reports and make documentation with effective presentation. - Learn to management and financial skills required for the execution of project.
CO3 → PO6 (3) CO3 → PO7 (3)	- Students can able to understand the impact of professional engineering solutions in societal and environmental contexts. - Students can be able to apply reasoning of contextual knowledge to assess societal, health and safety issues.

ACKNOWLEDGEMENT

Although a single sentence hardly suffices, we would like to thank almighty God for blessing us with his grace and taking our endeavour to a successful culmination.

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We extend our sense of gratitude to **Prof.V.R.Bagali , Head, Dept. of E&CE, JCE, Belagavi**, for extending support and cooperation which helped us in completion of the project work.

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ABSTRACT

“REAL TIME WATER ANALYSIS SYSTEM” is essential for ensuring safe and reliable water resources. This project presents an automated system that continuously monitors key water parameters, specifically turbidity and pH levels, using sensor data integrated with a machine learning model for accurate assessment and anomaly detection. The project mainly stress on the cost of the Instruments. The system processes real-time measurements to identify potential contamination or unsafe conditions and instantly notifies users through Gmail alerts whenever abnormal values are detected. By combining sensor technology, intelligent data analysis, and automated communication, the proposed solution enhances water quality monitoring with improved accuracy, efficiency, and responsiveness.

TABLE OF CONTENTS

CHAPTER 1 INTRODUCTION	11
1.1 MOTIVATION.....	11
1.2 PROBLEM STATEMENT	11
1.3 OBJECTIVE OF PROJECT.....	12
CHAPTER 2 HARDWARE DESIG.....	13
2.1 BLOCK DIAGRAM.....	13
2.2 PH METER.....	13
2.3 Turbidity Sensor.....	14
2.4 Arduino UNO.....	16
2.5 Django.....	17
CHAPTER 3 SOFTWARE DESIGN.....	18
3.1 PROGRAMMING LANGUAGE.....	18
3.2 DEVELOPMENT TOOLS.....	18
3.3 ARDUINO IDE	18
3.4 DESIGN	18
3.4.1 MAIN FLOW DIAGRAM.....	18
CHAPTER 4 APPLICATIONS.....	19
CHAPTER 5 RESULTS	23
CHAPTER 6 CONCLUSION AND FUTURE SCOPE.....	24
CHAPTER 7 REFERENCES	26

LIST OF FIGURES

Fig No.1: pH meter



Fig No.2: Arduino UNO



Fig No.3: Turbidity sensor



CHAPTER 1

INTRODUCTION

1.1 MOTIVATION

Real-time Water Quality Analysis offers a smart, efficient, and affordable way to ensure safe water by continuously monitoring key parameters like turbidity and pH levels. By integrating low-cost sensors with machine learning, the system can quickly detect abnormalities, learn patterns in water conditions, and predict potential issues before they become hazardous. Its automated Gmail notification feature keeps users informed instantly, enabling timely action and reducing the risk of consuming or using contaminated water. This cost-effective, data-driven approach not only improves water safety but also empowers communities and individuals with reliable, real-time insights into their water quality.

1.2 PROBLEM STATEMENT

Access to clean and safe water is often hindered by the lack of continuous and affordable monitoring systems, especially in resource-constrained regions. Traditional laboratory-based testing is time-consuming, expensive, and unable to provide immediate feedback on water quality changes. As a result, contamination events may go unnoticed until they pose serious health risks. The problem is further amplified by the absence of automated alert mechanisms that notify users in real time. Therefore, there is a need for a cost-effective system that can continuously monitor key water parameters—such as turbidity and pH—using low-cost sensors, analyze the data instantly, and notify users through email when water quality deviates from safe limits. Such a solution would help bridge the gap between affordability and timely water quality management.

1.3 OBJECTIVES OF PROJECT

To provide real-time tracking of critical water parameters such as turbidity and pH levels to ensure water safety.

To alert users immediately via Gmail notifications whenever water quality deviates from safe levels

To enable communities and individuals to make informed decisions about water usage based on real-time, reliable data.

1.4 DESIGN APPROACH

Firmware Design & Development

Hardware Interface Sensors:

Turbidity sensor measures water cloudiness.

pH sensor measures water acidity/basicity.

Optional: Temperature, TDS (Total Dissolved Solids), or ORP sensors for extended analysis.

Microcontroller: Arduino Uno to read sensor data.

Communication Module:

Arduino Uno uses serial port for sending data to a PC

Testing

a)Unit Testing

Test each sensor individually:Check if readings are within expected range.

Validate calibration formulas. Test Wi-Fi/email module separately:

Ensure successful connection to network.

Verify email delivery.

b) Integration Testing

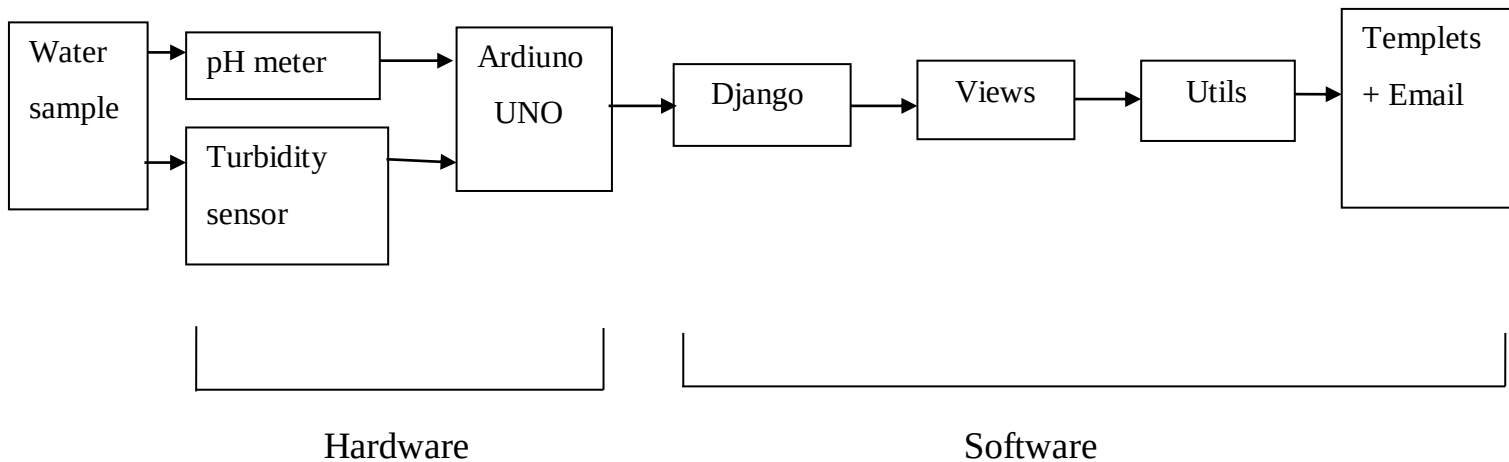
Combine all modules:

Sensor readings processing → alert → email.

CHAPTER 2

HARDWARE DESIGN

2.1 BLOCK DIAGRAM



2.2 EXPLANATION OF EACH BLOCK WITH PROPER NAMES AND THEIR FEATURES AS FOLLOWS

1) pH meter:



A pH meter is a scientific instrument used to measure how acidic or basic a solution is. It gives a numerical value called pH, which ranges from 0 to 14:

pH < 7 Acidic

pH 7 Neutral

pH > 7 Basic (alkaline)

Features:

A good pH meter provides precise readings (typically ± 0.01 to ± 0.05 pH).

This accuracy is crucial because even slight pH deviations can indicate contamination, chemical imbalance, or biological activity in the water.

The sensor gives stable readings within a few seconds.

This is essential for real-time monitoring, enabling the machine-learning model to track rapid changes in water chemistry.

Most pH sensors designed for projects come with:

- Analog output
- BNC connector
- Signal conditioning board

This makes integration with Arduino UNO or other controllers simple and stable.

2) Turbidity sensor:



A turbidity sensor is a device used to measure how clear or cloudy a liquid is. It detects the presence of suspended particles such as silt, clay, microorganisms, or other impurities in water. The measurement it provides is usually in NTU (Nephelometric Turbidity Units).

1. Inside the sensor, there is:

An LED (light source)

A photodiode (light detector)

2. Process:

The LED shines light into the water.

If the water is clear, most of the light passes through smoothly, and only a little is scattered.

If the water is cloudy, more light is scattered in different directions.

The photodiode measures the scattered light and converts it into a voltage.

3. Output:

Higher voltage = more turbidity (cloudy water)

Lower voltage = less turbidity (clear water)

Features:

Most turbidity sensors (like the SEN0189 or similar modules) can measure from 0 to 1000 NTU, allowing your system to detect anything from clear water to highly contaminated water.

The sensor is designed with an optical analog circuit that reduces signal disturbances. This ensures clean, stable readings, which is important for accurate ML training and automated alerts.

The turbidity sensor gives an analog voltage output, making it easy to interface with:

Arduino Uno

ESP32

NodeMCU

This suits your architecture perfectly, where Arduino reads sensor data and sends it to the ML model.

3) Arduino UNO:



Arduino Uno is a small open-source microcontroller board developed by Arduino. the board includes: a USB interface (to upload code from PC), a power jack (for external supply), a reset button, ground and power pins (5 V, 3.3 V, GND), and header pins for easy connection of sensors, modules, shields etc.

You write code (called a "sketch") in Arduino IDE, using a simplified version of C/C++. The IDE compiles and uploads the code to the UNO via USB. regentelectronics.com +1 Once uploaded, the microcontroller runs the code - it can read inputs from sensors (analog/digital pins), decide, and output signals e.g. turn on LEDs, drive motors, send serial data, etc.

Features:

This project requires reading analog values from:

- Turbidity sensor
- pH sensor

Any additional future sensors (TDS, temperature, etc.)

The Uno's 10-bit ADC (Analog-to-Digital Converter) converts real-world sensor signals into precise digital values, which is essential for accurate water quality analysis.

Arduino Uno has vast community support, libraries, and tutorials, making it easier to:

Interface sensors

Implement machine learning

Send Gmail alerts through IoT modules

This reduces development time and troubleshooting effort.

4) Django:

Django is a high-level Python web framework used to build secure and scalable web applications quickly. It follows the Model-View-Template (MVT) architecture and comes with many built-in features like authentication, admin panel, ORM, and routing. Django helps developers write clean, reusable code and speeds up development by handling common tasks automatically. It is widely used for websites, APIs, and large-scale applications.

CHAPTER 3

3.1 PROGRAMMING LANGUAGE

 Python

3.2 DEVELOPMENT TOOLS

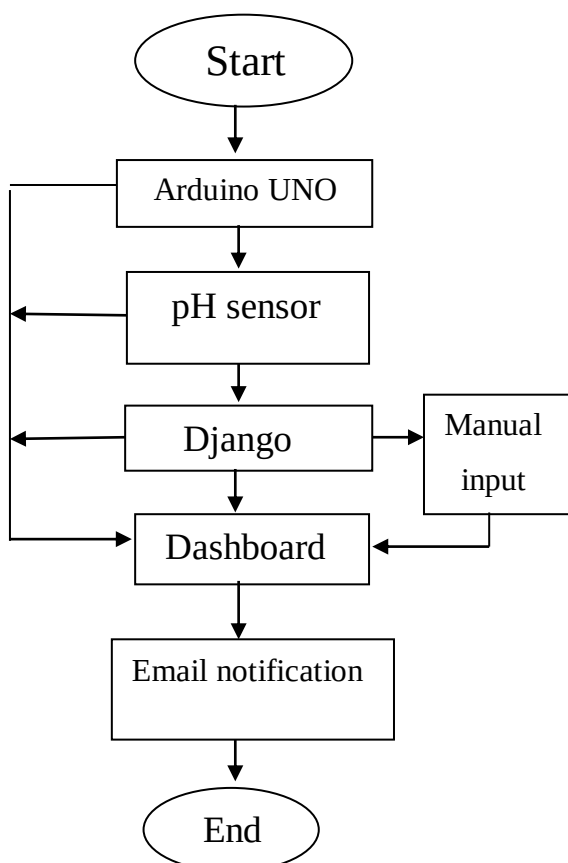
 Arduino SDK 1.6.9

3.3 ARDUINO IDE

Arduino software is needed to program Arduino boards and must be downloaded from the Arduino website and installed on a computer. This software is known as the Arduino IDE (Integrated Development Environment). Drivers must be installed in order to be able to program an Arduino from the Arduino IDE.

3.4 DESIGN

3.4.1 MAIN FLOW DIAGRAM



CHAPTER 4

APPLICATIONS

APPLICATIONS:

1. Household Drinking Water Monitoring

Your system can be installed in homes to continuously track pH, turbidity, and other parameters. Real-time alerts via Gmail help families detect contamination immediately and avoid unsafe water consumption.

2. Water Purification Plants

Treatment plants can use your setup to monitor the efficiency of filtration and purification stages. Machine learning can help predict when quality might dip, allowing staff to adjust the treatment process instantly.

3. Industrial Wastewater Monitoring

Industries can integrate this system to ensure the wastewater they release into the environment meets safety standards. Real-time analysis helps prevent environmental pollution and supports regulatory compliance.

4. Agriculture & Irrigation Water Testing

Farmers rely on good-quality water for crops. A real-time analysis setup can tell whether irrigation water has become too acidic, alkaline, or polluted, helping protect soil health and crop yield.

5. Aquaculture & Fish Farming

Fish and other aquatic species are extremely sensitive to changes in pH and turbidity. Continuous monitoring helps farm owners maintain healthy water conditions and avoid sudden fish mortality.

ADVANTAGES:

1.Can Predict Future Issues

Machine learning models recognize repetitive patterns and predict when contamination may occur. This helps in preventive maintenance rather than reacting to emergencies.

2. User-Friendly and Easy to Operate

With simple displays, email alerts, and intuitive dashboards, even non-technical users can monitor their water quality without any complexity.

3. Enhances Safety and Public Health

By ensuring safe drinking and usage water, the system contributes to healthier communities and reduces the burden on public health systems.

4.Environmentally Beneficial

Real-time monitoring prevents the unnoticed release of polluted water into natural sources, helping protect aquatic ecosystems.

5. Continuous and Instant Monitoring

The system keeps tracking water parameters like pH and turbidity every second or minute, providing a constant stream of accurate data. This eliminates the delays of traditional manual testing and ensures that unsafe water is detected immediately.

DISADVANTAGES:

1. Internet or Power Failure Affects Operation

Since the system depends on:

Internet for Gmail alerts

Power for continuous sensor reading

Any power outage or network failure may cause delays in monitoring or missed alerts, making the system less reliable in remote areas.

2. Limited Parameter Measurement

Systems measure only basic parameters like:

- PH
- Turbidity
- Temperature

But they cannot detect harmful contaminants like:

- Heavy metals
- Pesticides
- Microbial pathogens

Thus, the system cannot fully replace laboratory tests

3.Data Overload and False Alerts

Continuous monitoring generates a large amount of data. If the machine learning model or thresholds are not well-tuned, it may send:

- Too many Gmail alerts
- Alerts for minor, non-dangerous changes

This can cause alert fatigue, making users ignore important warnings.

4.Dependency on Power and Internet

Real-time monitoring requires continuous power and stable internet connectivity for data transfer and alert notification.

In rural or remote areas, power cuts or weak networks can disrupt real-time data transmission.

5. Sensor Accuracy Limitations

Low-cost sensors used for pH, turbidity, temperature, and other parameters may not always provide laboratory-level accuracy.

They can drift over time, require calibration, and sometimes show incorrect readings due to environmental factors like temperature changes or sensor aging.

CHAPTER 5

RESULTS

1. Accurate Detection of Water Parameters:

The system continuously monitored critical parameters such as pH levels and turbidity using low-cost sensors.

The readings were consistent and reliable, providing an accurate snapshot of water quality at any given time.

2. Automated Alerts for Abnormal Conditions:

The system detected abnormal changes in water quality and triggered automated Gmail notifications.

This feature ensured that users were immediately informed of potential hazards, allowing timely preventive actions.

3. Data Analysis and Prediction:

Integration with machine learning algorithms enabled the system to recognize patterns in water quality data.

It could predict potential contamination or abnormal trends, helping in proactive water management.

4. User-Friendly and Cost-Effective:

The project demonstrated a low-cost solution suitable for widespread use in communities, schools, and households.

The system empowered users with real-time, actionable insights into their water quality.

CONCLUSION:

The Real-Time Water Quality Analysis project demonstrates an effective and affordable approach to ensuring safe water by continuously monitoring critical parameters such as turbidity and pH. By combining low-cost sensors with machine learning, the system not only detects anomalies but also predicts potential water quality issues, enabling proactive measures. The integration of automated notifications ensures timely alerts, reducing health risks associated with contaminated water. Overall, this project highlights how technology can provide real-time, data-driven insights, promoting safer water consumption and empowering communities to make informed decisions about their water resources.

FUTURE SCOPE:

1. Integration with IoT and Cloud Platforms:

Expand the system to connect multiple sensors across different locations, sending data to the cloud for centralized monitoring, real-time dashboards, and advanced analytics.

2. Advanced Predictive Analytics:

Use machine learning models to predict long-term water quality trends, detect contamination sources, and provide early warnings before water becomes unsafe.

3. Mobile App Notifications:

Develop a mobile application to send instant alerts, show water quality trends, and provide recommendations for households, industries, and municipal authorities.

4. Additional Parameters Monitoring:

Include more water quality parameters like dissolved oxygen, electrical conductivity, heavy metals, and microbial contamination to make the system more comprehensive.

5. Automated Water Treatment Integration:

Link the system with automated water treatment solutions, like UV sterilizers or chemical dosing, to immediately correct water quality issues.

6. Community and Smart City Applications:

Deploy in public water systems, rivers, lakes, and smart cities to continuously monitor and ensure safe water supply for larger populations.

7. Cost Reduction and Miniaturization:

Further reduce the size and cost of the sensor units, making it accessible for rural areas and developing countries.

8. Data-Driven Policy Making:

Use the collected data to help governments and organizations make informed decisions about water management and environmental protection.

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